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Terraforming Mars in OpenGL

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Part I

Introduction

Terraforming Mars is a board game for 1 to 5 players. In it, the players control corporations which race to terraform the planet Mars. To achieve this, they must raise the temperature, form oceans and add oxygen to the atmosphere, and alongside these they create settlements, plant forest and much more.

The gameplay consist of generations, and each generation has three phases: research phase, action phase and production phase. In the research phase, players can choose from a few cards that represent projects contributing to the taming of Mars. In the action phase, players can use these cards, use actions on specific played cards, or perform standard projects. In the production phase, players get to replenish their resources based on their production values. Mars is represented by a hexagonal grid, on which players can place oceans, greeneries and cities.

The goal of this thesis was to implement this game with the ability to easily expand it: there are a variety of board variants, expansions, decks which the players can choose from. The other goal was to develop the application in a high performance, highly customizable environment supporting 3D rendering and models. Therefore, C++ and OpenGL were chosen for this project.

Part II

User documentation

1 Hardware and software requirements

1.1 Hardware

- 64-bit CPU with the x86 instruction set
- GPU supporting OpenGL 4.6
- 1 GB RAM
- mouse, optionally keyboard

1.2 Software

- Operating system: Windows 11
- GPU driver supporting OpenGL 4.6

2 Installation

The usage of the software does not require any form of setup or installation. After extracting the archive, the application can be run with the *view.exe* executable.

3 Usage of the application

The user can quit the application at any point by clicking the *Close* button of the window, or by pressing the following keyboard combination: **Alt + F4**

Fullscreen mode can be toggled at any point by pressing the key **F11** key.

3.1 Menu



Figure 1: The menu screen of the application with an ongoing game.

The menu screen has the following three buttons the user can interact with:

- **Continue:** this button is only shown if a game has already been started. Pressing this button brings back the player to the game screen with the ongoing game.
- **New Game:** pressing this button creates a new game and transitions the app to the game screen. If there was an ongoing game, it is lost forever.
- **Quit:** pressing this button exits the application, as if the *above detailed* methods were used. If there was an ongoing game, it is lost forever.

3.2 Game

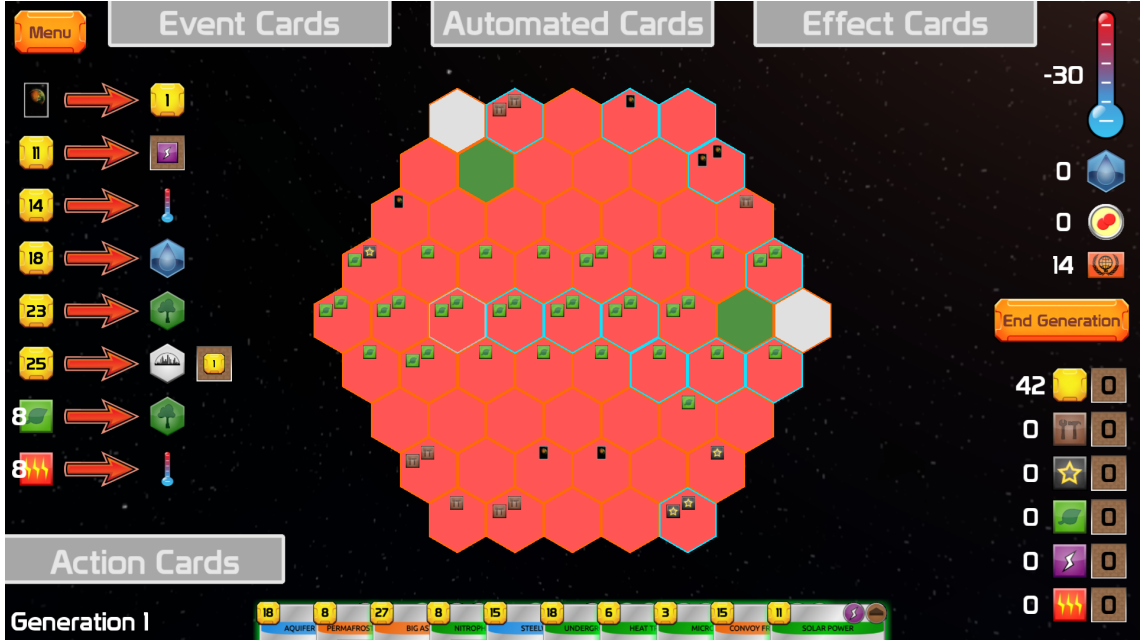


Figure 2: Game screen in the action state.

The player should be familiar with the *game rules*, in particular with the additions of the solo variant. Furthermore, these modifications are in effect:

- The player always controls the beginner corporation (10 cards for free and 42 Credits).
- Only about half of the cards from the basic set are included.

3.2.1 States

The game can be in different so called '*states*'. A state just means that certain actions may or may not be available to the player. The states are:

- **Action state:** This state is almost the same as the *Action phase* of the board game. The player can play cards from their hand, use standard projects or convert resources, use actions on their active cards and end the generation with the *End button*. The slight difference here is when using the *Sell patents* standard project, the game enters the *Sell state* where they can actually sell cards in their hand.
- **Sell state:** When using the *Sell patents* standard project, the game enters this state. In this state only the following actions can be performed:

-
- All cards in the *hand* glow red. The player has the option to drag out any card from their hand and play it. Playing the card sells it, discarding the card and giving one Credit to the player. For more detail about dragging out and playing cards, check the description of the *hand*.
 - The label of the *End button* is replaced. Clicking on it exits the *Sell state*, going back to the *Action state*.
- **Payment confirmation state:** When playing a card or using an action that can be bought with resources, the game will enter this state. In this state every action is disabled, and a window is shown where the player can confirm the amount of the resources they wish to spend. Confirming the payment advances the game, and returns to the *Action state*.



Figure 3: Payment confirmation window.

- **Placement confirmation state:** When playing a card or using a standard project or conversion that places a tile, the game will enter this state. The border of the tiles on *Mars* that are valid will be colored white, and the others will be dark. In this state every action is disabled, but the tiles of *Mars* are clickable, and clicking on a valid one places the tile there. A hint text informs the player what kind of tile they are placing currently. Confirming the placement advances the game, and returns to the *Action state*.



Figure 4: Placement confirmation. On the top, the tile to be placed is shown, and valid locations have a white border.

- **Research state:** In the *Action state*, clicking on the *End button* automatically performs the production phase, and after that the game will enter this state. In this state four cards will appear on the screen. The player needs to decide how many and which they want to purchase. Initially all of them are set for purchasing, and the player can toggle them between purchasing and discarding. This is backed up with visual cues. A hint text informs the player how much they are going to play for the selected cards. The label of the *End button* is replaced. Clicking on it confirms the research, and the game returns to the *Action state*.

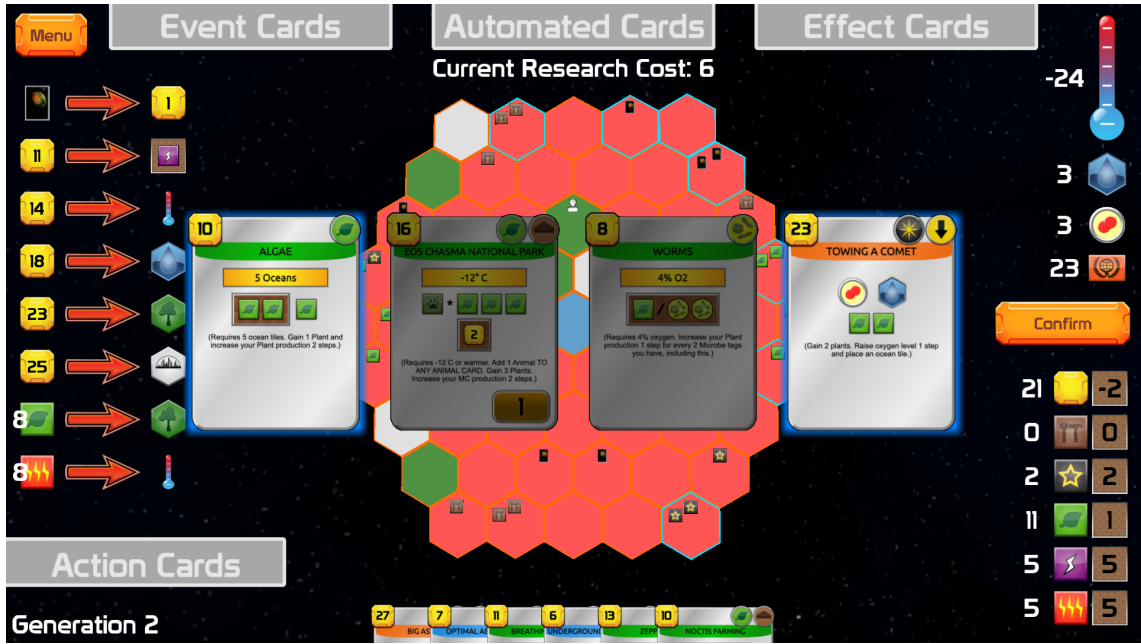


Figure 5: Research phase.

- **Post last generation state:** After ending the last generation and performing production phase, if the player has enough Plant resources to convert them to a greenery, the game will enter this state. Here, only the *Convert Plants to greenery* standard project can be used and the *End button* to finish the game.
- **Game over state:** After the last generation the game will enter this state. The game will display if the player has won or not, and the amount of victory points they have earned. Here, the player's only option is returning to the menu.



Figure 6: End of a game.

3.2.2 Elements

The game screen consists of the following visual elements:

- **Menu button:** Located in the top left corner. This button can always be clicked, and takes the player back to the menu screen. The ongoing game gets paused, and can be resumed by selecting the *Continue button* on the menu screen.
- **End button:** Located on the right side. It can have multiple behaviors, depending on the *state*.
 - In the *Action state*, the end button ends the generation.
 - In the *Sell state*, the end button exits the *Sell state*.
 - In the *Research state*, the end button confirms the research.
- **Standard projects and converting resources:** Located on the left side of the screen under the *Menu button*. When they can be used, hovering over them gives the visual cue of enlarging the row of the action. From top to bottom:
 - **Sell patents:** The application enters the *Sell state*.
 - **Power Plant:** For 11 Credits, you increase your Energy production one step.

-
- **Asteroid:** For 14 Credits, you raise temperature one step.
 - **Aquifer:** For 18 Credits, you place one ocean tile.
 - **Greenery:** For 23 Credits, you place one greenery tile.
 - **City:** For 25 Credits, you place one city tile and increase your Credit production one step.
 - **Convert Plants to greenery:** For 8 Plant resources, you place one greenery tile.
 - **Convert Heat to temperature:** For 8 Heat resources, you raise temperature one step.

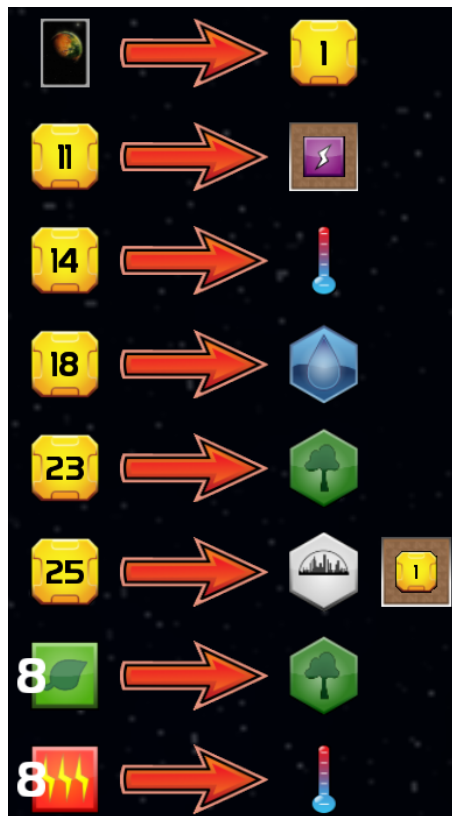


Figure 7: Standard projects and resource conversions.

- **Generation counter:** Located in the bottom left corner. Displays the current generation.
- **Global parameters:** Located on the upper right side of the screen. Shown parameters in order are: temperature, ocean count, oxygen, terraform rating. The three global parameter's indicators turn green when they have reached their respective goal.



Figure 8: Global parameters and terraform rating.

- **Resources:** Located on the lower right side of the screen. Consists of three columns: the first is the amount of resources, the second specifies the resource type and the third is the production output of said resource. The resources from top to bottom are as follows: Credit, Steel, Titanium, Plant, Energy, Heat. Values cannot be reduced below 0, except for Credit production which can be at a minimum of -5.

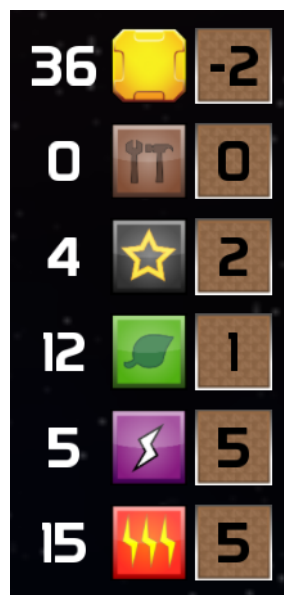


Figure 9: Resources. Notice how Credit generation is the only one that can go into negative, down to -5.

- **Hand:** Located at the bottom of the screen. Cards are by default hoverable: moving the mouse over a card enlarges it, showing the full card.

Some *states* allow the cards to be 'played'. In this case when a card is hovered, it can be dragged around with the mouse by pressing and holding the left mouse button. Two things can happen when the player releases the card: if the position of the mouse (and card) is above an invisible line, the card will get 'played', which is defined by the *state*. Otherwise, it will go back to the player's hand. There is always a visual indicator showing which one is about to happen.

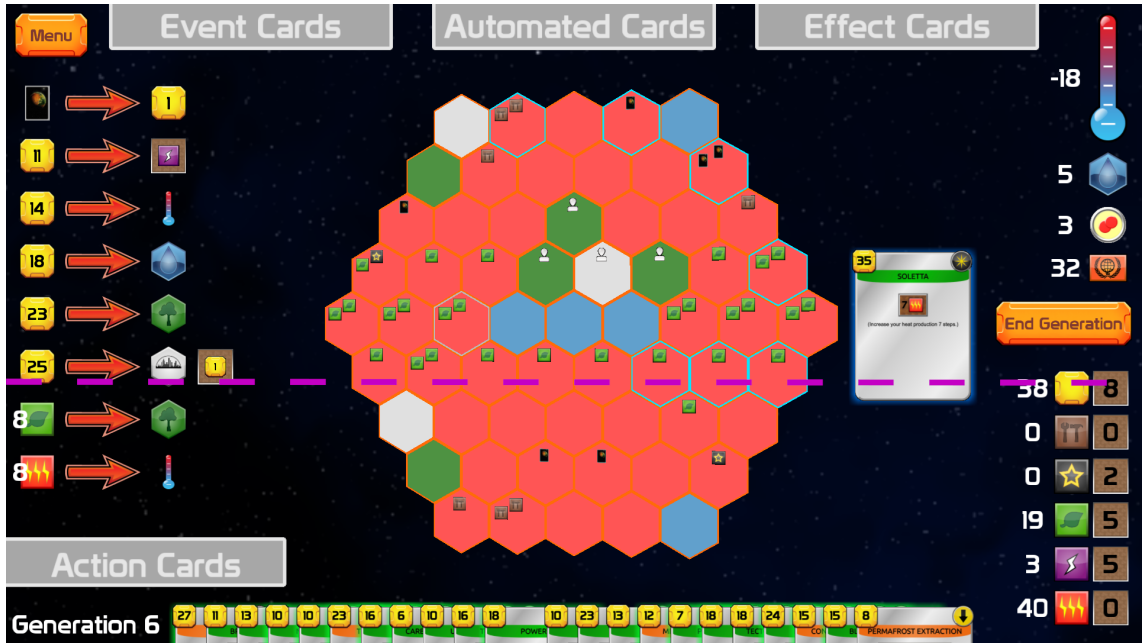


Figure 10: Approximate location of the above-mentioned invisible line.

- In the *Action state*, all cards that can be played will have a green glowing effect. If the mouse is over the invisible line, the card will have a blue glowing effect. The player can drag non-playable cards, but going over the invisible line automatically lets go of the dragged card.
- In the *Sell state*, all cards will have a red glowing effect. If the mouse is over the invisible line, the card will have a faded out effect. 'Playing' a card will sell it.



(a) Green highlight, usually indicating that the card can be played.



(b) Blue highlight, usually indicating that the card can be used or will be played.



(c) Red highlight, indicating that the card can be sold.



(d) Faded out card, usually indicating that the card cannot be used, will not be bought or will be sold.

- **Played card panels:** Four panels are located on the screen: action cards on the bottom left, event cards, automated cards and effect cards on the top. The panels open when they get clicked on, and close when any part of their body gets clicked, except the cards. They can also be toggled with the following keyboard keys: **U**, **I**, **O** and **P**. These can be opened at most times, but they do get closed automatically at certain events.

By default, only 8 cards can fit on a panel. If there are more out of the view, an arrow will be shown, which can be used to step one page in that direction. This can also be done with the **Left** and **Right** keys.

The action panel has further functionality: the card's actions can be used from here. If a card can be used, it will have a blue glowing effect. In this case, clicking on it will use the action of the card. If it was already used in the

generation, it will have a faded out effect. If it was not used in the generation but can't be used, it won't have any effects.

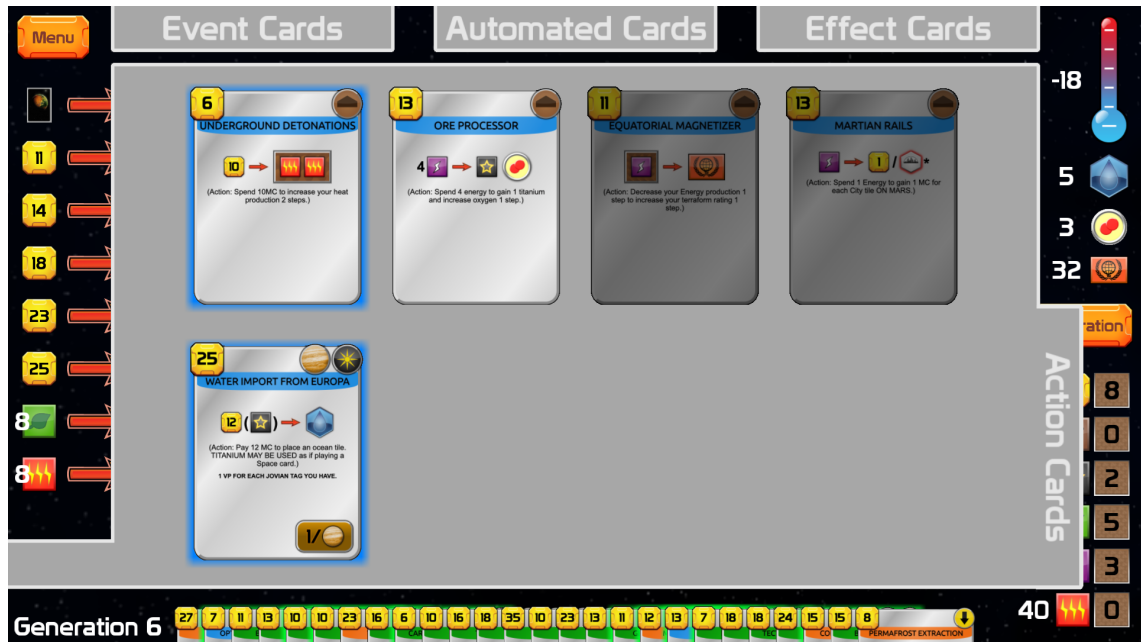


Figure 12: Action card panel. *Equatorial Magnetizer* and *Martian Rails* have been used this generation. While the others are still usable, *Ore Processor* cannot be used at this moment (the player does not have the required cost of 4 Energy).

- **Mars** Located in the center of the board. Empty tiles show their respective placement bonuses. Tiles owned by the player will have a player icon. Tiles reserved for oceans have a blue border. How placing tiles works is further explained under the section *Placement confirmation state*.

3.3 Debug tools

There are a few debug tools built in to the application. Both screens have a debug window that can be toggled by pressing **Ctrl + D**.

3.3.1 Menu

The debug window allows the user to set a seed for their game. Starting two games with the same seed means the random tiles will generate to the same place, and cards will come in the same sequence.

move this section to developer documentation

3.3.2 Game

The debug window allows the user to give themselves resources, resource production, raise their terraform rating, draw cards, raise temperature, raise oxygen and place ocean, greenery or city tiles.

Warning! The debug tool should only be used in the Action State. Usage in other states may cause the application to crash.

Pressing **F3** toggles the basic camera and the editorial camera. With the editorial camera, the user can use look around with the mouse and fly around with the following keys: **A**, **W**, **S**, **D**, **E**, **Q**¹.

¹A - left, W - forward, S - back, D - right, E - up, Q - down

Part III

Developer documentation

1 Developer environment

The program uses the **C++20** major version. Libraries were installed using the **vcpgk** package manager.

I developed the application on a laptop with **Windows 11**. I used Microsoft's IDE, **Visual Studio 2022** utilizing the **Desktop development with C++** workload and the **GLSL language integration** extension. It used version **10.0.22621.0** of the **Windows SDK**.

1.1 Utilized libraries

The following libraries and versions have been used for the application. Use this command to install the required dependencies using **vcpgk**:

```
.\vcpgk install sdl3 sdl3-image[jpeg,png] glew glm imgui[opengl3-binding,sdl3-bin
```

- **OpenGL:** (Apache License 2.0) - graphics rendering library.
- **OpenGL Extension Wrangler library (GLEW):** (MIT License) - library providing an interface to OpenGL.
- **OpenGL Mathematics:** (MIT License) - math functions, vectors, matrices, transformations, easing functions and more.
- **Simple DirectMedia Layer (SDL):** (zlib License) - window and input manager.
- **Dear ImGui:** (MIT License) - graphical user interface for debugging and tooling.
- **FreeType:** (The FreeType Project License) - library for font rendering.

1.2 Compilation

The project needs to be compiled with the following macros being defined: **SDL_MAIN_HANDLED**, **GLM_ENABLE_EXPERIMENTAL**, **GLM_FORCE_SWIZZLE**.

1.3 Runtime environment

Compiling the program in *Release* mode assembles the program in the *x64/Release/* directory. *view.exe* expects the assets, fonts and shaders to be in their

respective directories²: Visual Studio 2022 should take care of this during compilation. This directory can now be archived or moved, as it contains all the files required for running the application.

2 Specification

Menu

The player should be able to start a new game and exit the application. Furthermore, if there is an ongoing game, the user should be able to continue said game.

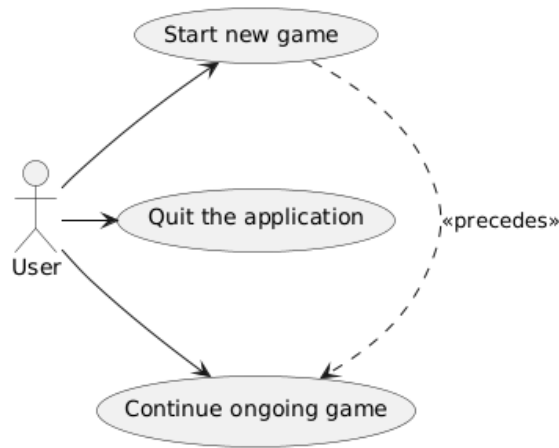


Figure 13: Use case diagram of the menu screen.

Game

The player should always have the option to leave the game. Available use cases change based on the *state* of the game.

²*assets/, fonts/, shaders/*

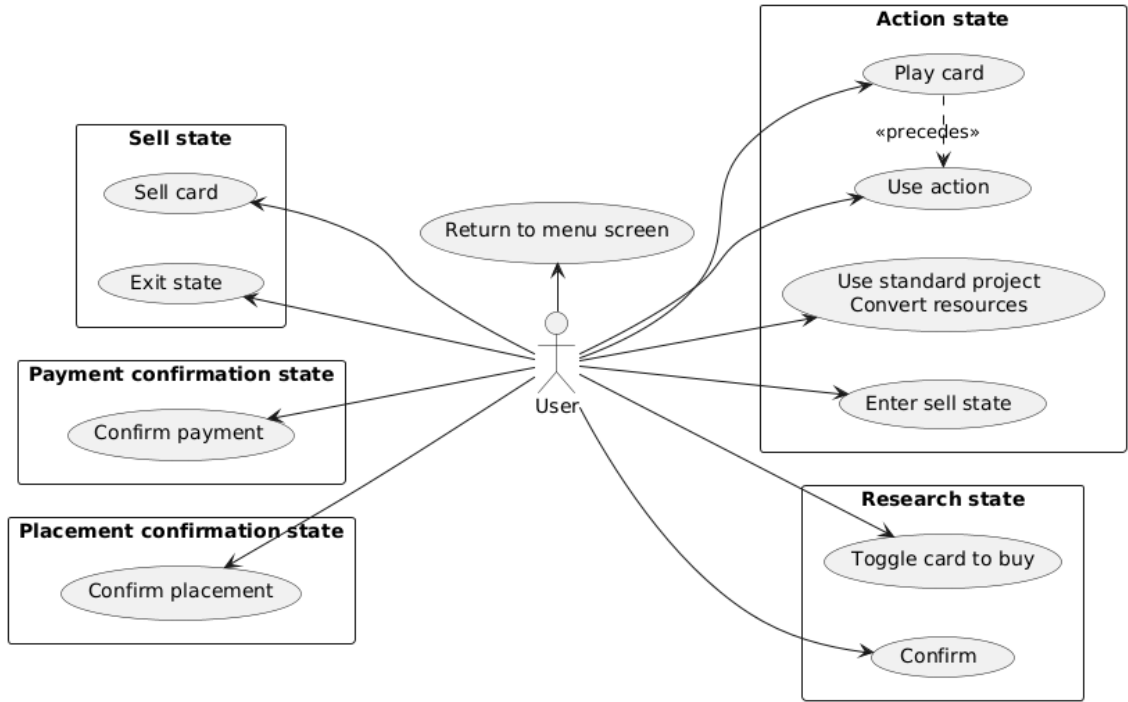


Figure 14: Use case diagram of the game screen.

Increasing a global parameter always increases the player's terraform rating.

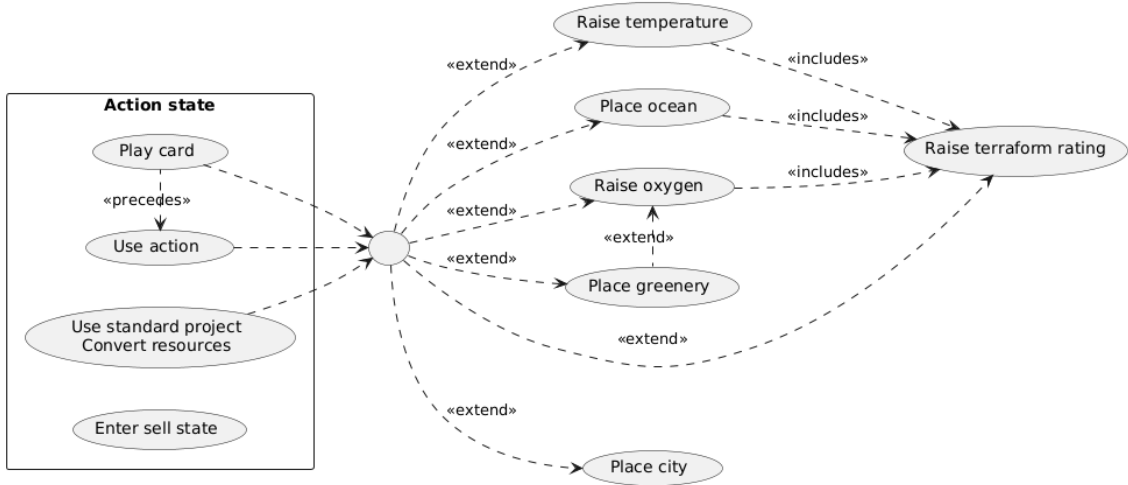


Figure 15: Detailed use cases of the Action state.

3 Architecture

The application has two layers: *Model* and *View*. Business logic is completely separated in the *Model*, and is platform independent. This means that separate frontends could be developed for this one backend.

The main aspect during development was extensibility: there are multiple boards

that can be played on, different decks using different cards, extensions which add new cards, new resources, or even new game features. In this documentation you can find multiple explanations for implementing new varieties.

View manages and renders the window, stores information for rendering and communicates with the *Model*. My implementation used OpenGL for rendering.

The *App* class holds together the application: it stores *View* instances, and injects instances of the *Model* into the corresponding *View* instance.

main.cpp is the entry point for the application. It manages the following responsibilities:

- Initializing submodules and libraries.
- Instantiate and initialize an *App* instance.
- Run the *render loop*. Here, it handles window events and forwards them to *App*, such as mouse and keyboard input. *App*'s *Update* and *Render* methods are called here.
- Clean up once the application is about to exit.

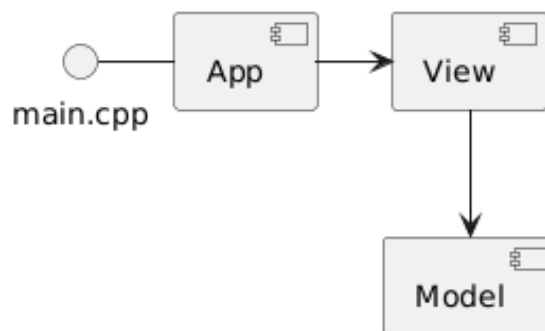


Figure 16: Package diagram of the application

Each file is self-contained, and only includes headers from the standard library other than the project. All enumerated values are *enum classes*.

There are some classes that have circular dependencies, for example *GameModel* and *GameModelState*. This was made possible with the following steps:

- Create *forward declaration headers* for the classes. This header should only have the declaration of it's class in it. Notice the naming of the file.

```
1 namespace model
2 {
3     class GameModel;
4 }
```

Code 1: game_model.fwd.hpp

- In the header files, the first lines should be including every **.fwd.hpp* file that is involved in the circular dependency. Include the rest of the header files as normal. DO NOT include the normal header file of other classes in the circular dependency.
- In the **.cpp* files, include the normal headers of all the classes needed.

This structure solves these circular dependencies, however, there are a few drawbacks:

- Since in the header files only the declaration of the other classes is present, no function calls can be made here. This means that inline functions cannot use the other classes.
- From the previous point, it is apparent that classes participating cannot be template classes, as these require all their definitions in their respective header files.

There is some leeway though: one header file from the two can include the other one. In this case, the structure of files and includes should still be upheld.

3.1 Model

Every code in this project is under the namespace *::model*.

3.1.1 The board

The *::model::boards* namespace contains the part of the code that implements the game board. Mars is made up of hexes, and to be more specific, the hexes have pointy-top orientation. Storing tables made up of hexes is not trivial, and there are

forrásmegjelölés multiple ways to implement it.

I choose to go with the axial coordinate system. Here every square has a *q* and an *r* coordinate. Increasing the *q* coordinate moves right horizontally, and increasing the *r* coordinate moves down, but instead of vertically, it moves on the line you get

by rotating the q axis 60° in the negative direction. This alignment matches with the usual layout in programming, where the y -axis is directed downwards.

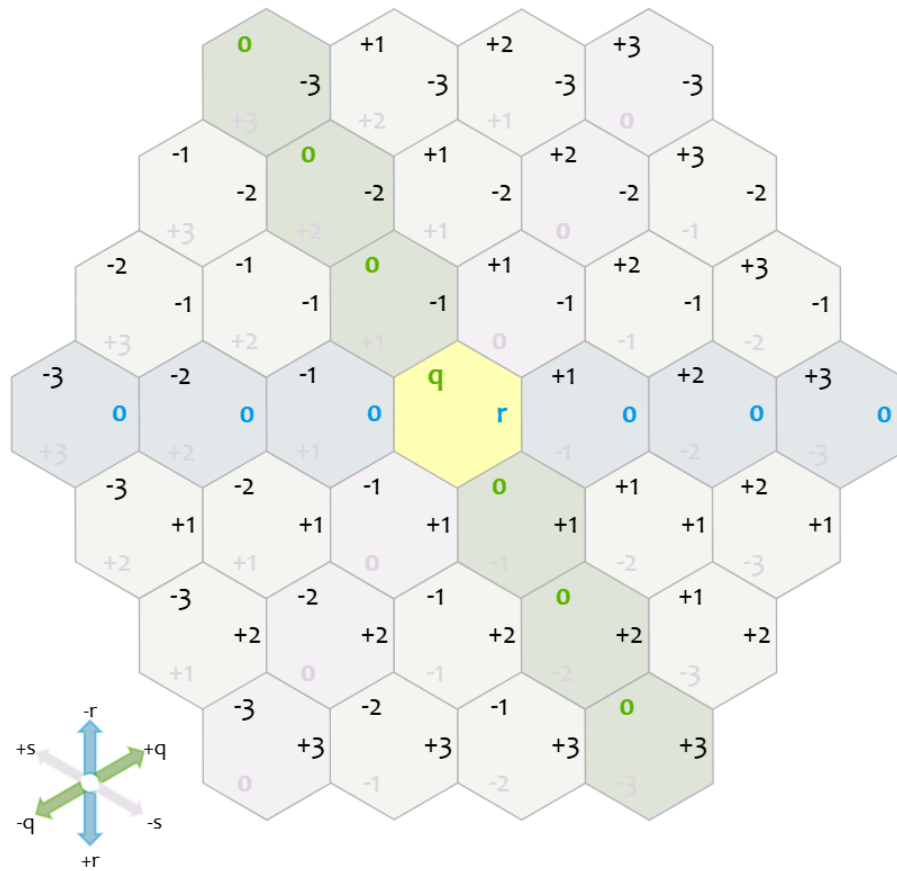


Figure 17: The axial coordinate system. Notice the third coordinate in a lighter color.

It is important to note, that the axial coordinates are derived from cube coordinates, which have a third coordinate, s . But since the equation $q + r + s = 0$ is true for all tiles, we can identify each square with just two coordinates, and calculate s if we need it.

The `::model::boards` namespace has one enumerated data type, `TileType`.

Tile

This class represents a single tile on the board. For convenience, they store their q and r positions, as well as their current type and owner. Oceans cannot be owned, but an empty tile can have an owner. Placement bonuses are also stored in each class and can be applied to the placer.

ezt itt ki kell jobban fejteni, vagy elég hogy az osztálydiagrammon rajta van?

Boards

To implement different boards in the application, I have decided to use the *bridge design pattern*, as

- it hides the underlying data structure in the form of the implementor,
- it enables different board layouts and implementations, as classes can derive from the implementor, each one representing a different board variant, *but* at the same time
- it makes the boards extensible, as classes can derive from the abstraction and implement functionality specific to them. Since we are deriving from the abstraction, all concrete implementors automatically gain this extended functionality.

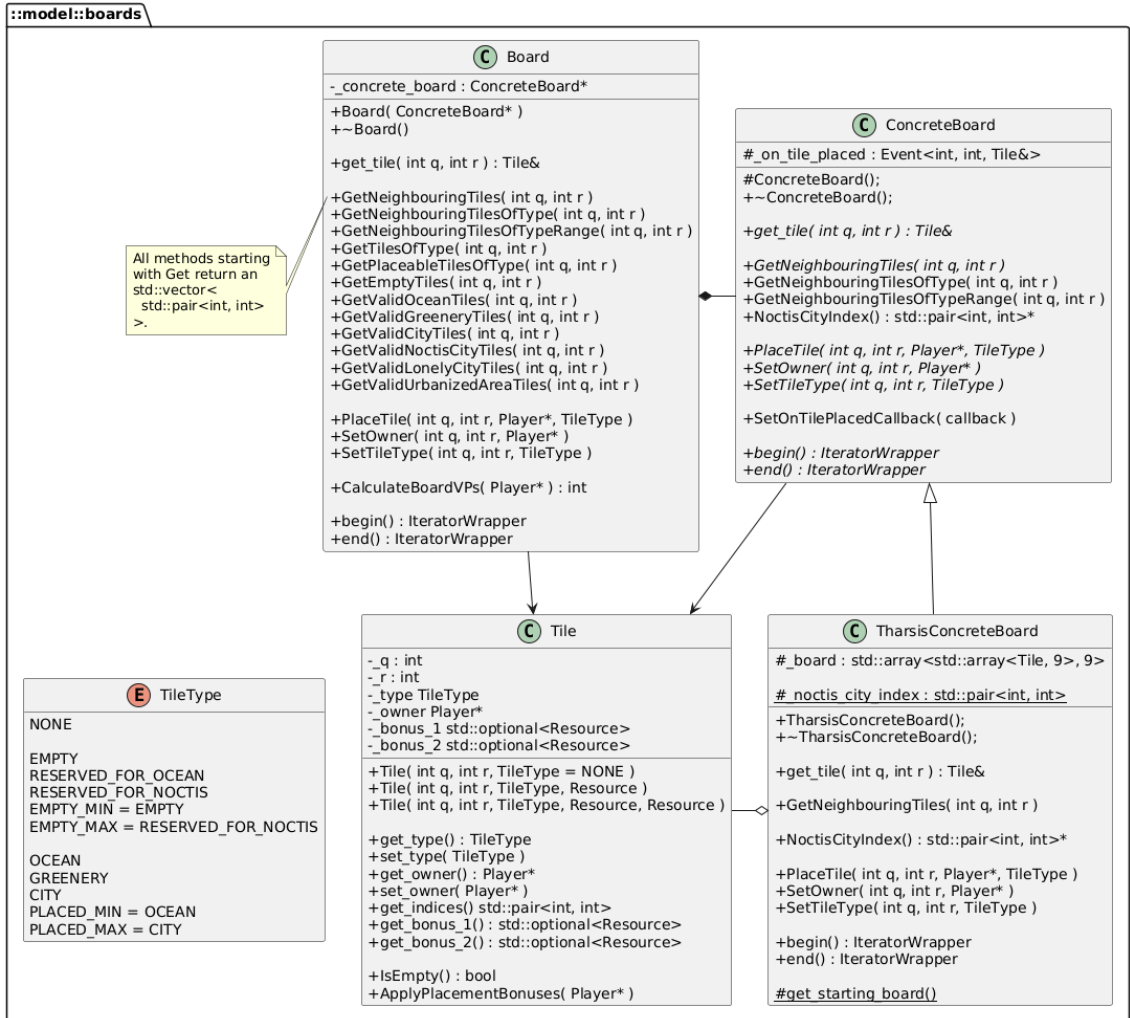


Figure 18: Class diagram of the `::model::boards` namespace.

As Figure 18 shows, `ConcreteBoard`'s methods are almost all virtual or abstract.

Board stores an instance of ConcreteBoard, and extends the interface of it with methods required for the game. A new board can simply be implemented by inheriting from ConcreteBoard. Here there are no such classes, but Board can also be derived from, to implement additional behavior, for example some functionality specific to an extension.

Since the game board as a whole is also hex shaped, the tiles do not make out a full rectangle. There are multiple solutions to storing a board, and each and every subclass of ConcreteBoard can implement their own.

For TharsisConcreteBoard, I choose the 2D array implementation shown on Figure 19. I made this decision because indexing is easy and fast compared to other solutions and the elements don't move, they stay in memory for a full game, so after the initial allocation only fields of the elements are modified. This also means that the unused memory is negligible.

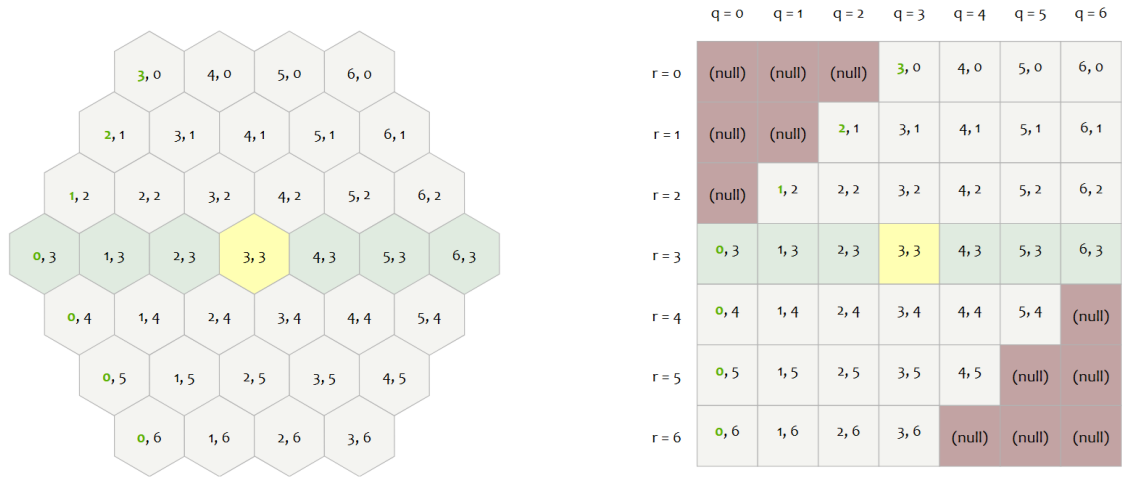


Figure 19: Example of memory layout when storing a board in a 2D array.

Iterators

Notice the *begin* and *end* methods in the class diagram on Figure 18. It was a requirement that boards could be iterated over. However, there were multiple obstacles I needed to overcome during the design and implementation of a universally usable iterator.

Since there could be any number of different board implementations, each one could potentially have a fundamentally different iterator. This sparked the interest to make an abstract Iterator base class, and derive iterators for every ConcreteBoard implementation.

But having `Iterator` as a base class meant that *begin* and *end* cannot simply return an `Iterator` instance because of object slicing³. The problem is, returning a pointer to an `Iterator` created by `new` does not sound good either: nobody expects a pointer when they call *begin* on a container and no one expects that they need to call *delete* on it. This would just be bad design. Furthermore, it would be a pointer: the semantics for using the iterator would be different, and you could not even use it for a range-based for loop.

My solution: I created a child class of `Iterator` called `IteratorWrapper`. It has a constructor that takes an `Iterator` pointer - any one of our concrete iterators - automatically deletes it when `IteratorWrapper` goes out of scope, and delegates every operation to the `Iterator` it is holding.

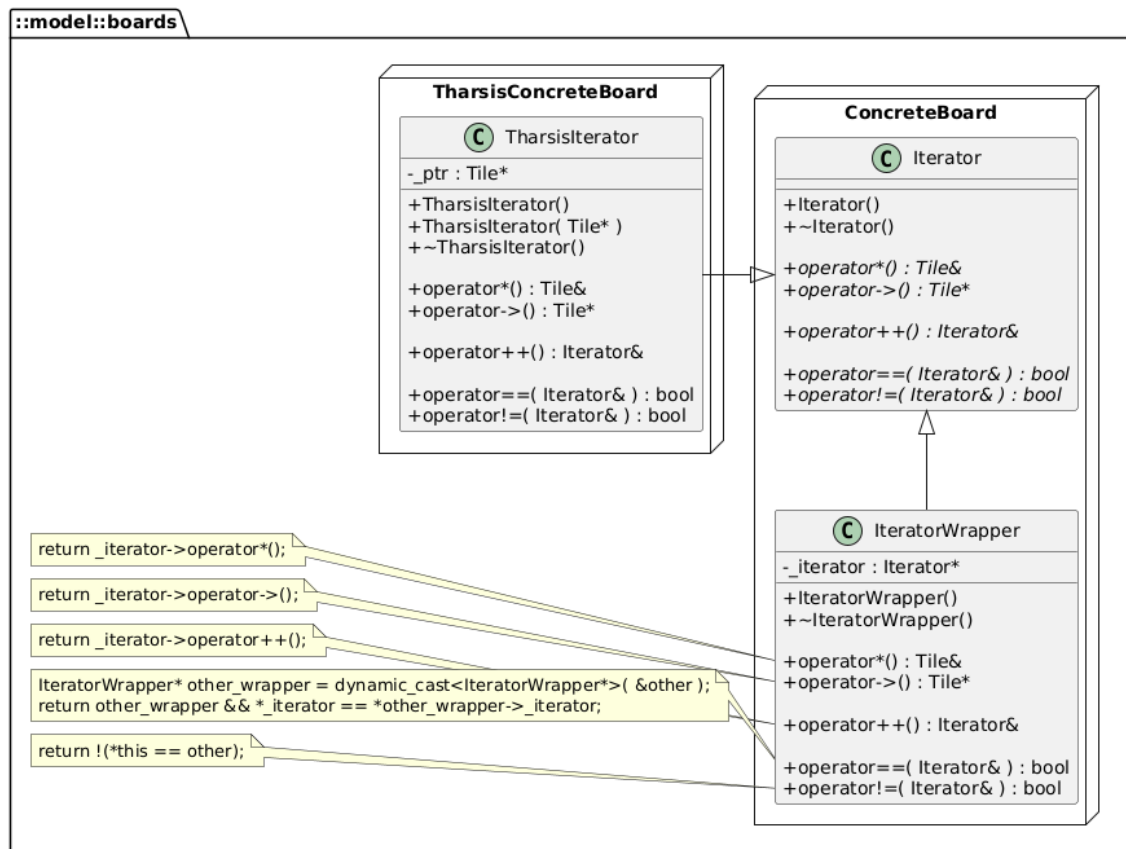


Figure 20: Class diagram of the iterators.

There are advantages and disadvantages of subclassing `IteratorWrapper` from `Iterator`. On one hand, it enforces any change in the `Iterator` interface to be updated in `IteratorWrapper`. On the other hand, someone may accidentally assign an instance

³The derived class needs more memory than its parent class.

of IteratorWrapper to an Iterator variable, and object slicing happens. However, this has a really low probability, as these classes are enclosed in the ConcreteBoard class.

3.1.2 Decks and cards

Part IV

Conclusion

Acknowledgements

Appendix A

Simulation results

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